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Full-Color Display Screen Secondary Development Interaction Protocol (JotuPix)

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1、 Introduction

1.1 Document Objective

1. Summarize the primary functionalities of the display screen.
2. Provides the Display Screen Interaction Protocol for developer use

1.2 Scope of Application (Optional)

Developer

2、 Device Overview

2.1 Device Introduction

Full-color display screen utilizing a flexible screen and potting process, allowing direct adhesion with seamless fit, no light leakage, lightweight, slim profile, and easy installation. Supports control via mobile APP and remote controller; the mobile APP can configure common functions such as text, animations, doodles, and music beat response. Supports multiple built-in content types (text, doodles, animations) and local microphone beat response, integrating both information display and entertainment functions. Functionalities may vary between different display screen models.

2.2 Function Summary

1. Supports multiple built-in content types (including text, doodle, and animation)
2. Supports text setting functionality
3. Supports configuration of doodle and animation
4. Music and microphone rhythm with multiple music beat effects, also supports local microphone functionality
5. Supports multiple display modes and text color modes

Note: Functional differences may exist between different types of display screens

3、 Power Supply Requirements

Display screen power supply: DC 5V; operating voltage must not exceed this range to avoid device malfunction or damage

Note: Power requirements vary with display screen resolution; please refer to the display screen specifications

4、 Data Format

Data transmission must conform to a fixed data format; otherwise, the device will not recognize the data.

1	2	3	4	5				n+3	n+4
0x01	n		data						0x03
Start Byte	Data Length		Data						End Byte

➤ Start Byte

Fixed value of 0x01, occupying one Byte; Within the interactive data, except the Start Byte which is 0x01, no other data can assume this value.

➤ Data Length

Consists of two Bytes, indicating the length of the current transmitted data segment, excluding the Start Byte, End Byte, and Data Length fields. Each length unit represents one Byte, with the High Byte preceding. For example, if the data length is 0x0017, the second Byte corresponds to the data 0x00, and

the third Byte corresponds to 0x17.

➤ Data

Data to be transmitted

➤ End Byte

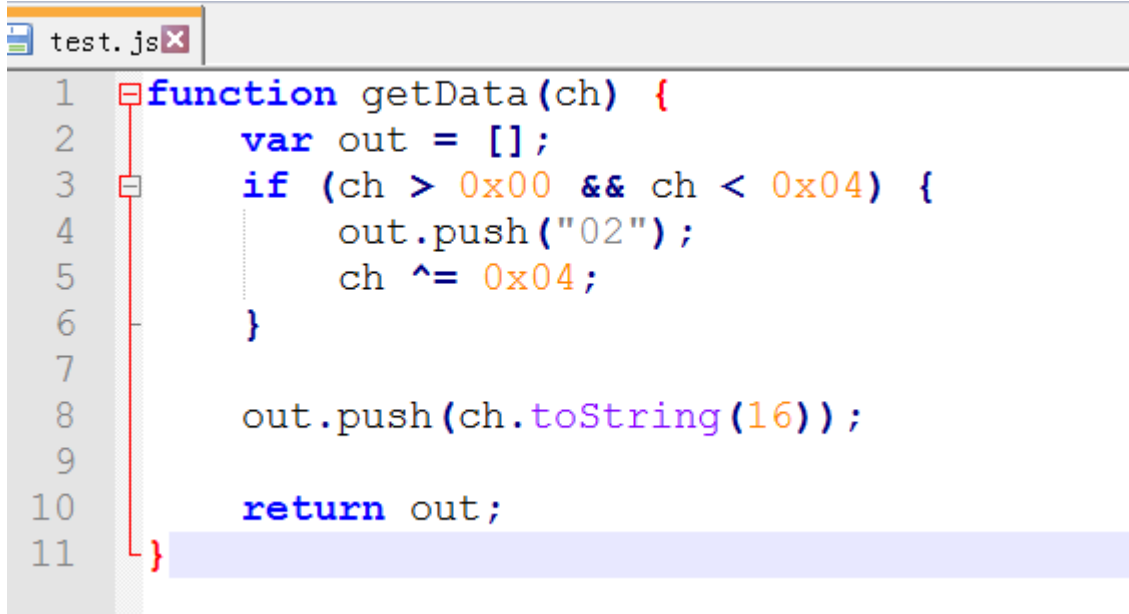
Fixed value set to 0x03, occupying one Byte; In interactive data, except for the start bit being 0x03, no other data can have this value.

➤ Data mandatory conversion

All data except the start bit and end bit must undergo mandatory conversion before transmission, the conversion requirements are as follows:

If any Byte in the transmitted data is greater than 0x00 and less than 0x04, a Byte with value 0x02 must be inserted before it, and the Byte is XORed with 0x04 before transmission. The inserted 0x02 Byte is not included in the Data Length.

Reference:



```

1 function getData(ch) {
2     var out = [];
3     if (ch > 0x00 && ch < 0x04) {
4         out.push("02");
5         ch ^= 0x04;
6     }
7
8     out.push(ch.toString(16));
9
10    return out;
11 }

```

Example:

For example, it is necessary to transmit a data segment as follows: 0x00 0x01 0x05

Data Length: 0 x0003, since the high byte of the data length 0x00 is not within the required conversion range, it remains unchanged, whereas 0x03 requires conversion to 0x02 0x07

The final transmitted data is: 0 x01 0x00 0x02 0x07 0x00 0x02 0x05 0x05 0x03

Example code for data format processing:



app_commpt.c



app_commpt.h

5、 Data Packetization Processing

Due to excessively long data and the limited RAM capacity of the device, the device cannot accept overly long

data in a single transmission, therefore, to enhance transmission reliability, certain transmitted data require packetization.

package Data Format:

Package Type	Data Total Length	Package ID	Current Package Data Length	Current Package Data	Checksum
1byte	4byte	2byte	2byte	nbyte	1byte
Package Type	Refers to the total length of the original data before segmenting into packages. (If compression is applied, this refers to the compressed total length; otherwise, it is the original data length)	For each new content set, Package ID starts at 0, where 0 represents the initial package of the transmission, incrementing sequentially thereafter.	Maximum data length per package. Default is 1 024.		The XOR of 0 x00 and all preceding data bytes in sequence, yielding the final value.

Note:

1. Each package's data must include the message type. Equivalent to Transmitted Data: type+package.
2. The transmitted packet data (type+package) must be converted according to the [Data Format](#) for the device to correctly recognize it.
3. Upon transmitting each packet data (type+package), the device will return a response, and the APP must handle the response message.
4. Only after receiving the response can the APP initiate the next operation (such as transmitting the next packet, retransmitting data, or displaying transmission failure).

Acknowledgment:

<type> Message Type, occupying one byte.

< ack > Transmission result of the corresponding packet ID, structured as follows:

Package Type	Package ID	error_code
1byte	2byte	1byte
Default value is 0	0~0xFFFF	Possible values are as follows: 0 x00 Transmission Successful Other Transmission Failure (Retransmit All Data)

During APP transmission, timeout must be considered. If no response is received within 5 seconds after sending each packet, it indicates a transmission timeout and retransmission (restart transmission) is required. If retransmission exceeds 3 attempts and still fails, it indicates the current doodle data setting has failed.

Note: The response data must also conform to the data format requirements; therefore, the response data must be parsed accordingly.

6、 Data Compression

Before data packet transmission, data compression must be performed once to reduce the volume of transmitted data, facilitating faster completion of data transfer. This employs the lzss compression algorithm, which achieves a high compression ratio for data with significant redundancy. Since doodle, animation, and text data on the display screen typically exhibit high redundancy, the compression ratio generally ranges from 60% to 85%. Reference implementation is provided in the lzss.js file.

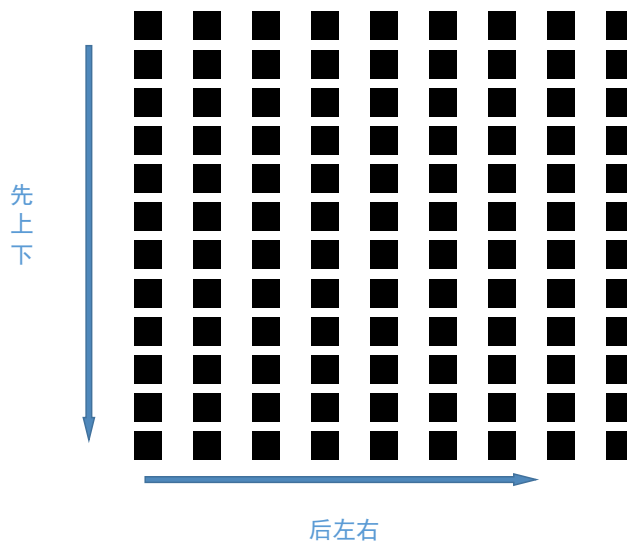


7、 Grayscale Introduction

Grayscale refers to the number of adjustable brightness levels for the color components supported by the display screen. For instance, a full-color 16-level grayscale means that the device uses RGB components, each with 16 brightness levels. Thus, the value range for each color component is: 0~ 15. By combination, the total number of colors that can be represented is $16*16*16= 4096$.

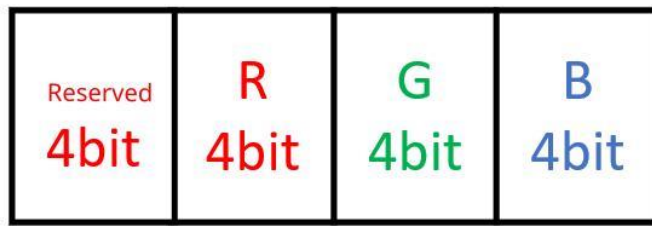
The APP supports 256 grayscale levels; therefore, data conversion is required, with each component value divided by 16.

8、 Data scanning method



The scanning method used is: first vertical (top to bottom), then horizontal (left to right), with the high bit preceding.

9、 Data Arrangement Method

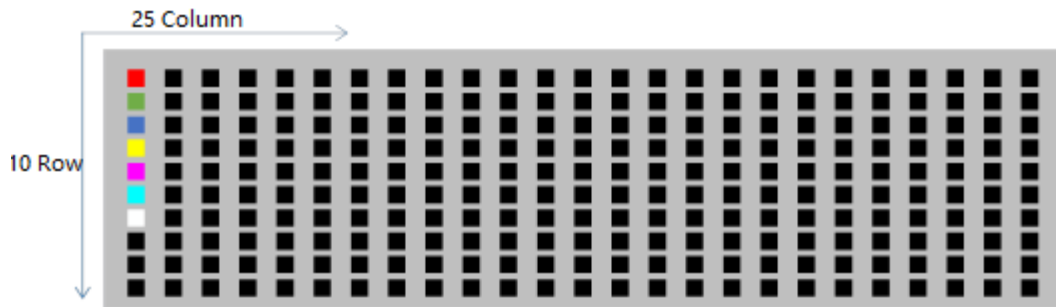


bit15

bit0

All data consists of three RGB components, with colors encoded in RGB444 format, where each point's color is represented by 2 bytes (a total of 16 bits), and each color component occupies 4 bits. Bits 0 to 3 represent the blue component, bits 4 to 7 represent the green component, bits 8 to 11 represent the red component, and bits 12 to 15 are reserved and set to zero. For example, for a full-color screen with a resolution of 16 by 64, one frame of data equals $16 \times 64 \times 2 = 2048$ bytes.

Example:



Taking the above figure as an example, each point is a pure color with a maximum brightness value of 0x0E, and the representation method for each point in every column is:

First column:

Point 1, Red: Red component 0x0E, Green component 0x00, Blue component 0x00, corresponding data value 0x0E00

Point 2, Green: Red component 0x00, Green component 0x0E, Blue component 0x00, corresponding data value 0x00E0

Point 3, Blue: Red component 0x00, Green component 0x00, Blue component 0x0E, corresponding data value 0x000E

Point 4, Yellow: Red component 0x0E, Green component 0x0E, Blue component 0x00, corresponding data value 0x0EEE

5th point, Purple: Red component 0x0E, Green component 0x00, Blue component 0x0E, corresponding data 0x0E0E

6th point, Cyan: Red component 0x00, Green component 0x0E, Blue component 0x0E, corresponding data 0x00EE

7th point, White: Red component 0x0E, Green component 0x0E, Blue component 0x0E, corresponding data 0x0EEE

8th point~10th point, Black: Red component 0x00, Green component 0x00, Blue component 0x00, corresponding data 0x0000

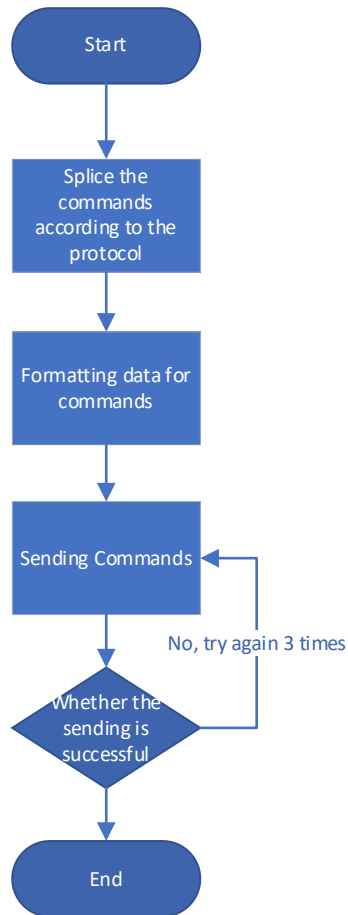
Data of the first column:

0E 00 00 E0 00 0E 0E 00 0E 0E 00 EE 0E EE 00 00 00 00 00 00

All other column data are zeros

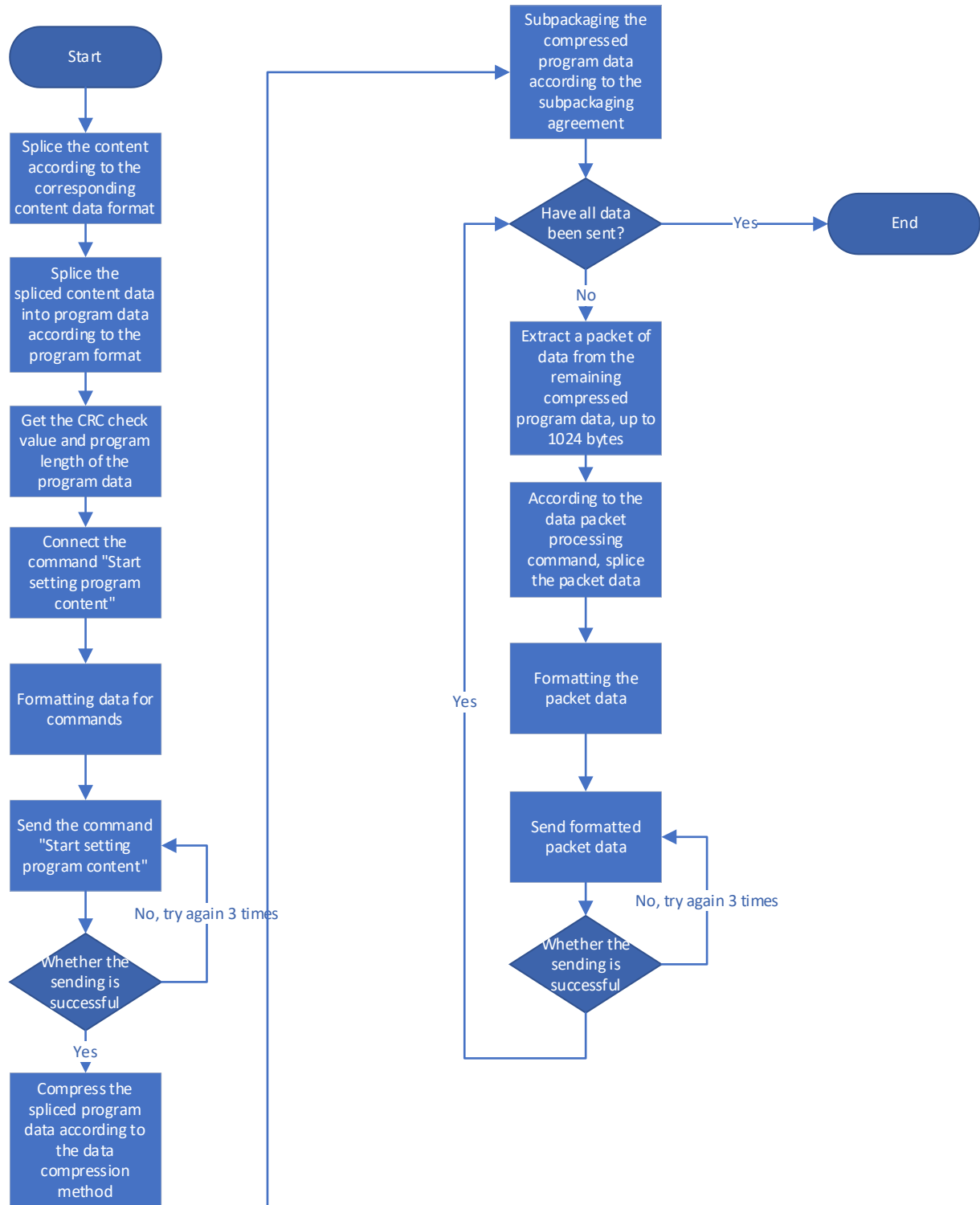
10、 Data Transmission Process

10.1 General Data Transmission Process



10.2 Transmission Process with Compression and Segmentation

Using the program transmission process as an example, the OTA upgrade process is identical.



11、 Interaction Protocol

11.1 Mobile Music Rhythm Mode (Microphone Mode)

Command Description: When the application is playing music or is in microphone mode, it extracts the music rhythm, converts it into bar data, and sends it to the module.

Command Format:

Direction	Type	Params	Response
APP→LIGHT	0x01	<type><param>	None

Data Description:

<0x01> Message type, occupying one byte, and always the first byte of a Bluetooth Data packet segment

<type> Music mode type, one byte, values refer to the appendix 'Music Mode' section

< param> Parameters corresponding to the music mode, 8 bytes: <Height of column 1> <Height of column 2> ... <Height of column 8>

Note: Music-driven data transmission should not be too rapid, with an interval of 50 ms to 100 ms.

11.2 Begin setting program content

Command Description: This command initiates the transmission of program content and must be invoked prior to setting the program content.

Command Format:

Direction	Type	Params	Response
APP→LIGHT	0x02	Reference data description	<0x02><Ack>

Data Description:

<002> Message type, occupies one byte, and must always be the first byte of a Bluetooth Data packet.

Parameter description:

Parameter Name	Length	Description	Remarks
<pro-data-crc>	4byte	CRC checksum value of bytes 3 and 2 of the current program data (computed before compression)	CRC 32 checksum value calculated for the program data
<pro-data-len>	4byte	Length of the current program data (uncompressed length)	
<pro-index>	1byte	Order position of the current program in the program list to be sent, starting from 0.	Value: 0~n-1.
<pro-num>	1byte	Total number of programs to be sent	Value range: 1~ n Current supported number of programs, to be retrieved from the device.
<play-cnt>	1byte	Original program playback count, no longer supported; default is 1.	Value: 1
<reserved>	7byte	Reserved byte(s)	Default value is 0
<compress>	1byte	Compression flag	0- Program is compressed

			1- Program is not compressed
<pro-group>	1byte	Program group category	
<group-param>	nbyte	Parameters for different program categories	

Description of program group categories and parameters:

1. General program class parameters, including text, graffiti, animation, etc.

Parameter Name	Length	Description	Remarks
<pro-group>	1byte	Program Group Category	Fixed to 0
<group-param> Composed of the following parameters:			
<play-type>	1byte	Playback Mode	0x00: Play by count 0x01: Play by duration
<play-param>	4byte	Playback Parameters	When playback mode is 0 x00, it represents the number of times, Value range: 1~255, Default is 1 time. When playback mode is 0 x01, it represents playback duration in seconds, with a maximum not exceeding 1 day, Default is 30 minutes.

2. Parameters for Real-time Image Program Type:

Parameter Name	Length	Description	Remarks
<pro-group>	1byte	Program Group Category	Fixed to 1
<group-param> Composed of the following parameters:			
< reserved >	8byte	Reserved bytes, all set to 0	

Note: For real-time image program types, the Program Content must contain only real-time image data; otherwise, display anomalies will occur.

Acknowledgment:

<0x02> Message type, occupies one byte, and must be the first byte of a Bluetooth Data segment.

< Ack> Response Status, 0- Success, no local copy exists; program content needs to be transmitted; 1 - Local copy exists; no need to transmit the program content, directly confirm successful sending and proceed to transmit the next program; Others - Failure.

Note:

- 1) Transmission of program data may only commence when the response status indicates success.
- 2) Upon successful receipt of this command, the device will first check locally for a copy of the program; if found, retransmission is unnecessary.
- 3) If compressing the program is inconvenient for the host computer, compression may be omitted; however, the compression flag must be set to 1. This will increase the data volume and extend transmission time; please make trade-offs based on actual conditions.

For the CRC 32 checksum algorithm, please refer to the appendix titled 'CRC32 Checksum Algorithm'

11.3 Set Program Content

Command Description: This command is used to set the content of each program. Each program's content may include multiple elements such as text, graffiti, and animations.

Command Format:

Direction	Type	Params	Response
APP→LIGHT	0x03	Reference data description	<0x03><ack>

Data Description:

<0x03> Message type, occupies one byte, and must be the first byte of a Bluetooth data packet

Parameter description:

Parameter Name	Length	Description	Remarks
<reserved>	8byte	Reserved byte(s)	Default value is 0
<pro-content-num>	1byte	Number of content elements included in the program	Value range: 1 ~n
<reserved>	1byte	Reserved byte(s)	
<pro-cnt-data-1>	nbyte	Data of the corresponding content (contents rendered earlier correspond to a lower rendering layer level Z)	Content may be text, graffiti, or animations. Refer to the respective content data format.
.....			
<pro-cnt-data-n>	nbyte	Data of the corresponding content (contents rendered earlier correspond to a lower rendering layer level Z)	Content may be text, graffiti, or animations. Refer to the respective content data format.

Note: < Params > This data must first be compressed, then subjected to packet segmentation. For details, please refer to Data Compression and Data Packet Segmentation and Processing.

Acknowledgment:

Refer to Packet Segmentation Response Data [Data Packet Segmentation and Processing](#).

11.3.1 Text Content Data Format

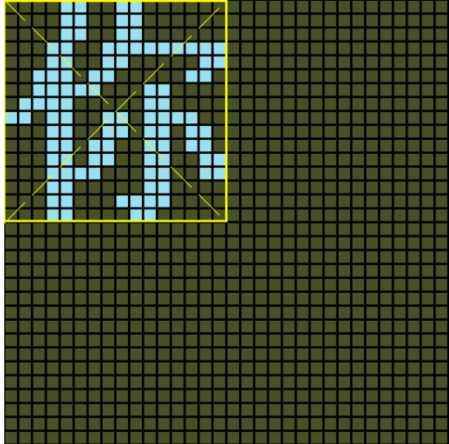
Parameter Name	Length	Description	Remarks
<content-size>	4byte	Total length of all data in this content segment	Includes the 4 bytes of <content-size> itself
<content-type>	1byte	Specifies the type of this content	Fixed value: 0 x01
<reserved>	7byte	Reserved byte(s)	Default value is 0
<blend-type>	1byte	Blending mode of this content when displayed with other layer levels	Value range: 0- Blending (the higher Z-level layer blends with the lower Z-level layer based on transparency) 1- Overlay (the higher Z-level layer directly overwrites the lower Z-level layer)

<show-start-column>	2byte	Start column of this content display, equivalent to x-coordinate	Value range: 1 ~ Number of display screen rows minus the number of rows required for one text character (must be able to display at least one text character)
<show-start-row>	2byte	Start Row for This Content Display, equivalent to y-coordinate	Value range: 1 ~ (the number of display screen columns minus the width of one character in columns) (at least sufficient to display one character)
<show-width>	2byte	Display Width of This Content	Value range: width of one character ~ display screen width
<show-height>	2byte	Display Height of This Content	Value range: height of one character ~ display height
<show-mode>	1byte	Display Mode	Value reference: see the "Appendix - Display Mode" section
<show-speed>	1byte	Display Speed (speed corresponding to the display mode)	Value: 1 ~255 (the larger the value, the faster the speed)
<stay-time>	1byte	Dwell time (duration after one full screen display)	Value: 0 ~255 seconds (this parameter is invalid in continuous scrolling modes)
<movespace>	2byte	Movement interval	This parameter is effective only in continuous leftward or rightward scrolling modes; otherwise, it defaults to the current display content width. Movement interval value range: 0~ display content width (default is display content width)
<text-num>	2byte	Number of characters	Supports up to 150 characters (Note: one emoticon counts as two characters)
<text-all-wide>	4byte	Total width occupied by all text	
The following content is the data array for each character, arranged in order			
<text-wide>	1byte	Width occupied by the first character	
<text-type>	1byte	Type identifier of the first character	0 - denotes monochrome text, 1 - denotes multicolor text (emoticon)
<text-data>	nbyte	Display data of the first character	If it is monochrome text, only display data is required If it is multicolor text (emoticon), the character data is represented using the graffiti method.
.....			
<text-wide>	1byte	Width occupied by the Nth character	
<text-type>	1byte	Type identifier of the Nth character	0 - denotes monochrome text, 1 - denotes multicolor text (emoticon)
<text- data>	nbyte	Display data of the Nth character	If it is monochrome text, only display data is required If it is multicolor text (emoticon), the

			character data is represented using the graffiti method.
--	--	--	--

Note: Since characters can be of different sizes, the position and size of the displayed content here should be adjusted according to the size of the character.

For example, as shown in the illustration below:



Within a 32x 32 display area, when selecting a 16x 16 character size, the text content cannot completely fill the entire area. Therefore, the visible height should not be 32, as this would cause display disorder; it should instead be set to 16. If the user also selects an alignment method, the display position of the text content must be adjusted correspondingly.

Text Content Remarks:

1. Methods to Acquire Text Display Data

There are two methods to acquire text display data:

- 1) The first method is to directly provide a dot matrix font library. However, this method uses a fixed font size and does not support arbitrary scaling, although it allows modification and optimization of individual characters.
- 2) The second method scales the text to a specified size based on a vector font library, then rasterizes it to produce the corresponding display data. This method supports arbitrary scaling of text size.

2. Example:

1) Font Library Approach

The font library uniformly employs the Unicode (UCS-2) standard, supporting commonly used Simplified Chinese, Traditional Chinese, Korean, Japanese, Latin scripts, and others. It covers nearly all characters and symbols globally, making it a reasonable choice as a font library. Unicode (UCS-2) encoding uses two bytes to represent one character, allowing it to accommodate over 60,000 characters in total. The font sizes in the font library vary, resulting in different byte counts for the display content of each character.

Byte count = ((fontlibSize + 7) / 8) * fontlibSize (fontlibSize refers to the dot matrix font size; “/” denotes integer division, retaining only the integer part)

The dot matrix data is then read from the font library file according to the character's Unicode encoding.

Position calculation: offset = character's Unicode encoding * byte count

Subsequently, read the specified number of bytes equivalent to the byte count. There is no need to differentiate languages; unified conversion to Unicode encoding is sufficient.

For example, the Unicode12 dot matrix font library uses a font size of 12. Each character occupies 24 bytes. The addressing formula is: character Unicode encoding multiplied by 24, and the number

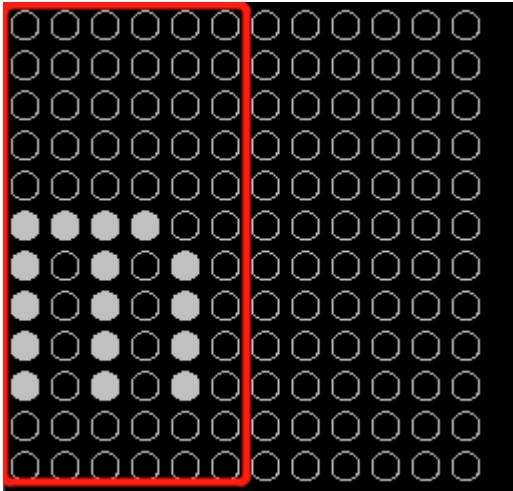
of bytes read per access is 24.

The font library file is a binary file arranged according to the Unicode encoding sequence of characters' display data. Therefore, it can be accessed as a byte stream. Note that file operation interfaces differ across platforms.

3. Handling of blank spaces in character display

Some characters occupy only a few columns of dot matrix data, while the rest are blank, causing excessive inter-character spacing. Therefore, redundant blank dot matrix data must be removed, leaving only one column of blank space as spacing between characters.

For example, the dot matrix data for m is as follows:



(Also applicable to cases where the text is right-aligned)

There are a total of 12 columns of data, with only 5 columns containing valid data, plus one blank column as a separator. Therefore, the valid data constitutes only 6 columns. The remaining blank columns must be removed.

The display data corresponding to m is as follows:

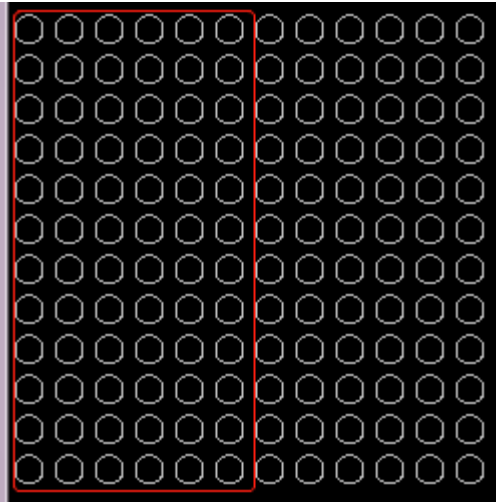
```
unsigned char data[] = {
    0x07,0xc0,0x04,0x00,0x07,0xc0,0x04,0x00,0x03,0xc0,0x00,0x00,
    0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,
};
```

This corresponds to traversing backward from the end of the data array, grouping every two bytes as one unit of data, until encountering one of these two bytes that is non-zero (indicating that the column is not empty, i.e., traversing up to the 4th column of m (starting from column 0)). Then, the data from columns 0 to 4 are copied, and one empty column is appended at the end of the array to serve as a separator. The resulting m data is:

```
{0x07,0xc0,0x04,0x00,0x07,0xc0,0x04,0x00,0x03,0xc0,0x00,0x00}
```

4. Handling the situation where data read from the font library consists entirely of zeros

For example: space characters; in such cases, half of the data must be discarded, retaining only half.



When encountering empty data, only half of the empty data should be retained.

11.3.2 Graffiti Content Data Format

Parameter Name	Length	Description	Remarks
<content-size>	4byte	Total length of all data in this content segment	Includes the 4 bytes of <content-size> itself
<content-type>	1byte	Specifies the type of this content	Value fixed at 0x02
<reserved>	7byte	Reserved byte(s)	Default value is 0
<blend-type>	1byte	Blending mode of this content when displayed with other layer levels	Value range: 0- Blending (the higher Z-level layer blends with the lower Z-level layer based on transparency) 1- Overlay (the higher Z-level layer directly overwrites the lower Z-level layer)
<show-start-column>	2byte	Start column of this content display, equivalent to x-coordinate	Value range: 0 to (Display Screen Width - 1) (values outside this range will not be rendered)
<show-start-row>	2byte	Start Row for This Content Display, equivalent to y-coordinate	Value range: 0 to (Display Height - 1) (values outside this range will not be rendered)
<show-width>	2byte	Display Width of This Content	Value range: 1 to Display Screen Width (start column + width must not exceed display screen width)
<show-height>	2byte	Display Height of This Content	Value range: 1 to Display Screen Height (start row + height must not exceed display screen height); height affects the byte count per column of data
<show-mode>	1byte	Display Mode	Value reference: see the "Appendix - Display

			Mode" section
<show-speed>	1byte	Display Speed	Value Range: 1 ~255
<stay-time>	1byte	Dwell time (duration after one full screen display)	Value: 0 ~255 seconds (this parameter is invalid in continuous scrolling modes)
<data-size>	4byte	Graffiti Total Length of Dot Matrix Data	
<data>	nbyte	Graffiti Dot Matrix Display Data Corresponding to	

Data method refers to the data arrangement method.

11.3.3 Animation Content Data Format

Animation data consists of consecutive frames of graffiti data, with a fixed time interval between each frame, operating on the same principle as a GIF image.

Parameter Name	Length	Description	Remarks
<content-size>	4byte	Total length of all data in this content segment	Includes the 4 bytes of <content-size> itself
<content-type>	1byte	Specifies the type of this content	Value fixed to 0x03
<protocol-ver>	1byte	Protocol Version Number	Fixed to 0x01
<reserved>	6byte	Reserved byte(s)	Default value is 0
<blend-type>	1byte	Blending mode of this content when displayed with other layer levels	Value range: 0- Blending (the higher Z-level layer blends with the lower Z-level layer based on transparency) 1- Overlay (the higher Z-level layer directly overwrites the lower Z-level layer)
<show-start-column>	2byte	Start column of this content display, equivalent to x-coordinate	Value range: 0 to (Display Screen Width - 1) (values outside this range will not be rendered)
<show-start-row>	2byte	Start Row for This Content Display, equivalent to y-coordinate	Value range: 0 to (Display Height - 1) (values outside this range will not be rendered)
<show-width>	2byte	Display Width of This Content	Value range: width of one character ~ display screen width
<show-height>	2byte	Display Height of This Content	Value range: height of one character ~ display height
<rev>	1byte	Reserved byte(s)	Default value is 0
<frame-num>	2byte	Number of Frames	Maximum of 40 Frames
<frameEveInterval>	nbyte	Independent Display Time Set for Each Frame	Display time for each frame, occupying 2 bytes each, arranged sequentially.

<frameData>	nbyte	Frame Data Set (corresponding to frameNum frames)	The size of each frame's data must match the size of the display area, and frameData must be an integer multiple of a single frame. Different device types use different data composition methods.
-------------	-------	---	--

11.3.4 Border Content Data Format

Parameter Name	Length	Description	Remarks
<content-size>	4byte	Total length of all data in this content segment	Includes the 4 bytes of <content-size> itself
<content-type>	1byte	Specifies the type of this content	Value fixed at 0x04
<reserved>	7byte	Reserved byte(s)	Default value is 0
<blend-type>	1byte	Blending mode of this content when displayed with other layer levels	Value range: 0- Blending (the higher Z-level layer blends with the lower Z-level layer based on transparency) 1- Overlay (the higher Z-level layer directly overwrites the lower Z-level layer)
<show-start-column>	2byte	Start column of this content display, equivalent to x-coordinate	Value range: 0 to (Display Screen Width - 1) (values outside this range will not be rendered)
<show-start-row>	2byte	Start Row for This Content Display, equivalent to y-coordinate	Value range: 0 to (Display Height - 1) (values outside this range will not be rendered)
<show-width>	2byte	Display Width of This Content	Value range: 1 to Display Screen Width (Start Column + Width exceeding the display screen width will be clipped)
<show-height>	2byte	Display Height of This Content	Value range: 1 to Display Screen Height (Start Row + Height exceeding the display screen height will be clipped); height affects the byte count per column of data.
<border-type>	1byte	Border Display Effects	0- Reserved 1- Clockwise Rotation 2- Counterclockwise Rotation 3- Blinking 4- Static
<border-speed>	1byte	Border Change Speed	Value: 1 ~255 (the larger the value, the faster the speed)
<border-height>	1byte	Border Content Height	Value range: 1 ~ Up to half of the display height. For a display screen with a typical height of 16 rows, the border height can be

			at most 1 row; otherwise, it will obscure excessive display content.
<data-size>	2byte	Total Length of Border Data	
<data>	nbyte	Border Display Data	Data composition is determined by the data modulus and data arrangement method.

Only one segment of border data needs to be provided; the device will automatically repeat this segment of display content.

Example 1:



To implement a border of height 1 row arranged as 'red-black' as shown in the figure above (the subsequent points repeat and do not require data), follow the data and color arrangement rules.

Red: 0x 0F,0x00

Black: 0 x00,0x00

Therefore

<border-height> is set to: 1

<data-size >is set to: 4

<data >is: 0x 0F,0x00,0x00,0x00

Example 2:



To implement a border with a height of 2 rows as illustrated above,

based on the data bit extraction method, vertical first then horizontal, the resulting bit layout is:

Red Purple Black Black (subsequent repeated data need not be provided)

According to the data layout rules, the color representation is as follows:

Red: 0x 0F,0x00

Purple: 0x 0F,0x0F

Black: 0 x00,0x00

Accordingly:

<border-height> is set to: 2

<data-size >is set to: 8

< data > is: 0x 0F,0x00, 0x 0F,0x0F, 0x 00, 0x00, 0x 00, 0x00

11.3.5 Text color data format

This feature applies exclusively to text and requires that this data segment be placed prior to the Text Content Data. In fact, the text color data and Text Content Data are linked together; otherwise, the setting is invalid. Only one of the two text effects, either the colorful text effect or the custom text effect, can be selected; simultaneous selection is not permitted.

11.3.5.1 Vivid Text Effects

Parameter Name	Length	Description	Remarks
<content-size>	4byte	Total length of all data in this content segment	Includes the 4 bytes of <content-size> itself
<content-type>	1byte	Specifies the type of this content	Value fixed to 0x05
<reserved>	7byte	Reserved byte(s)	Default value is 0
<show-start-column>	2byte	Start column of this content display, equivalent to x-coordinate	Value matches the Start Row in the Text Content Data
<show-start-row>	2byte	Start Row for This Content Display, equivalent to y-coordinate	Value matches the Start Column in the Text Content Data
<show-width>	2byte	Display Width of This Content	Value matches the Display Width in the Text Content Data
<show-height>	2byte	Display Height of This Content	Value matches the Display Height in the Text Content Data
<textcolor-type>	1byte	Text Color Effect	0- Reserved 1- Horizontal Scroll (Left, Right) 2- Horizontal Static (None; default direction is 0) 3- Vertical Scroll (Up, Down) 4- Vertical Relative Scroll (Toward Center, Toward Both Sides) 5- Jump Transition (None; default direction is 0) 6- Horizontal Overlay (Left, Right) 7- Horizontal Diagonal Scroll (Left, Right) 8- Windmill Rotation (Left, Right)
<textcolor-speed>	1byte	Text Color Effect Transition Speed	Value: 1 ~255 (the larger the value, the faster the speed)
<textcolor-dir>	1byte	Text Color Effect Display Direction	0- Left 1- Right 2- Up 3- Down 4- Toward Center 5- Toward Both Sides Supported directions vary among different effects
<reserved>	1byte	Reserved	Default value is 0
<data-size>	2byte	Color Data Length	
<data>	nbyte	Color Data	The color layout is constructed according to the data arrangement method

Example:

To set the text color to a static horizontal rainbow effect, the seven rainbow colors are: Red, Yellow, Green, Cyan, Blue, Purple

According to the color layout rules:

Red: 0x 0F, 0x 00

Yellow: 0x 0F, 0xF0

Green: 0x 00, 0xF0

Cyan: 0x 00, 0xFF

Blue: 0x 00, 0x0F

Purple: 0x 0F, 0x0F

Therefore

<data-size > is: 0x 0C (Data Length is 12)

<data > is set to: 0x0F,0x00,0x0F,0xF0,0x00,0xF0,0x00,0xFF,0x00,0x0F,0x0F,0x0F

< textcolor-type> is set to: 2

< textcolor-dir> is set to: 0

11.3.5.2 Custom text effects

Custom text effects enable users to define the color of each individual character.

Parameter Name	Length	Description	Remarks
<content-size>	4byte	Total length of all data in this content segment	Includes the 4 bytes of <content-size> itself
<content-type>	1byte	Specifies the type of this content	The value is fixed at 0 and 0x06.
<reserved>	5byte	Reserved byte(s)	Default value is 0
<movespace>	2byte	Movement interval	The value corresponds to the movement interval as specified in the text content data.
<show-start-column>	2byte	Start column of this content display, equivalent to x-coordinate	Value matches the Start Row in the Text Content Data
<show-start-row>	2byte	Start Row for This Content Display, equivalent to y-coordinate	Value matches the Start Column in the Text Content Data
<show-width>	2byte	Display Width of This Content	Value matches the Display Width in the Text Content Data
<show-height>	2byte	Display Height of This Content	Value matches the Display Height in the Text Content Data

<show-mode>	1byte	Display Mode	The value corresponds to the display mode as specified in the text content data.
<show-speed>	1byte	Display Speed (speed corresponding to the display mode)	The value corresponds to the display speed as specified in the text content data.
<stay-time>	1byte	Dwell time (duration after one full screen display)	The value corresponds to the dwell time as specified in the text content data.
< reserved >	1byte	Reserved	Default value is 0
<text-num>	2byte	Number of characters	Supports up to 150 characters.
<text-all-wide>	2byte	Total width occupied by all text	
<text-eve-wide>	nbyte	A collection of the width occupied by each character.	Each character's width is represented by one byte; the length of this data segment = < text-num >.
<text-eve-color>	nbyte	A collection of the color for each character.	Text colors are arranged according to the specified data arrangement method. [If the character is multicolor (emoticon), it may be represented by the default red.]

Example:

For example, this setting includes 7 characters with the text content '君同科技柔性屏', each character size is 32, and the text colors are respectively 'Red, Green, Blue, Yellow, Cyan, Purple, and White.'

According to the color layout rules:

Red: 0x 0F, 0x 00

Green: 0x 00, 0xF0

Blue: 0x 00, 0x0F

Yellow: 0x 0F, 0xF0

Cyan: 0x 00, 0xFF

Purple: 0x 0F, 0x0F

White: 0x 0F, 0xFF

Purple: 0x 0F, 0x0F

Accordingly:

<text-num>: 00 07

< text-all-wide >: 00 E0 (Total width of all characters)

<text-eve-wide>: 20 20 20 20 20 20 20

<text-eve-color>: 0x0F, 0x00, 0x00, 0xF0, 0x00, 0x0F, 0x0F, 0xF0, 0x00, 0xFF, 0x0F, 0x0F, 0x0F, 0xFF

Note: If colors are all represented using graffiti mode, the data volume will be 17 times larger. In actual use, text is the primary element supplemented by emoticons. Therefore, the color representation methods for text and

emoticons are differentiated here.

11.3.6 GIF Animation Data Format

Send the GIF image directly to the device for display.

Note: This data format can only be used when the program content contains a single GIF image or when the GIF display area does not overlap with other content display areas; otherwise, use the animation content data format.

Parameter Name	Length	Description	Remarks
<content-size>	4byte	Total length of all data in this content segment	Includes the 4 bytes of <content-size> itself
<content-type>	1byte	Specifies the type of this content	Value fixed at 0 x0C
<reserved>	7byte	Reserved byte(s)	Default value is 0
<blend-type>	1byte	Blending mode of this content when displayed with other layer levels	Value range: 0- Blending (the higher Z-level layer blends with the lower Z-level layer based on transparency) 1- Overlay (the higher Z-level layer directly overwrites the lower Z-level layer)
<reserved>	1byte	Reserved byte(s)	Default value is 0
<show-start-column>	2byte	Start column of this content display, equivalent to x-coordinate	Value range: 0 to (Display Screen Width - 1) (values outside this range will not be rendered)
<show-start-row>	2byte	Start Row for This Content Display, equivalent to y-coordinate	Value range: 0 to (Display Height - 1) (values outside this range will not be rendered)
<show-width>	2byte	Display Width of This Content	Value range: 1~Display Screen Width
<show-height>	2byte	Display Height of This Content	Value range: 1~Display Screen Height
<gif-data-size>	4byte	Total length of GIF image data	
<gif-data>	nbyte	GIF image data content	Data exceeding the display area will be cropped and not displayed. The device does not perform scaling.

Note: GIF image file size must not exceed: Display Screen Dot Matrix Height x Display Screen Dot Matrix Width x 2 x 40 (Bytes)

11.3.7 Real-time image data format

Real-time rendered image content. This image content is not retained after power-off and will be lost after reboot. It is intended for scenarios requiring display of real-time changes.

Note: Real-time images can only be used as a standalone program and cannot be combined with other program content.

Parameter Name	Length	Description	Remarks
<content-size>	4byte	Total length of all data in this content segment	Includes the 4 bytes of <content-size> itself
<content-type>	1byte	Specifies the type of this content	Value fixed at 0 x0E
<reserved>	7byte	Reserved byte(s)	Default value is 0
<show-start-column>	2byte	Start column of this content display, equivalent to x-coordinate	Value range: 0 to (Display Screen Width - 1) (values outside this range will not be rendered)
<show-start-row>	2byte	Start Row for This Content Display, equivalent to y-coordinate	Value range: 0 to (Display Height - 1) (values outside this range will not be rendered)
<show-width>	2byte	Display Width of This Content	Value range: 1 to Display Screen Width (start column + width must not exceed display screen width)
<show-height>	2byte	Display Height of This Content	Value range: 1 to Display Screen Height (start row + height must not exceed display screen height); height affects the byte count per column of data
<reserved>	8byte	Reserved byte(s)	Default value is 0
<data-size>	4byte	Corresponding image Total length of the dot matrix data	The size of the image data corresponds exactly to the image dimensions.
<data>	nbyte	Corresponding image dot matrix display data	

Note: Do not exceed the transmission rate of the serial port or BLE, as this will cause frame loss.

11.4 Set Display Brightness

Command Description: This command is used to set the display brightness.

Command Format:

Direction	Type	Params	Response
APP→LIGHT	0x04	<bn>	<0x04><bn>

Data Description:

<0x04> Message type, occupying one byte, and must be the first byte of a Bluetooth Data segment.

<bn> Display Brightness, occupying one byte, value range: 0 ~255

Acknowledgment:

<0x04> Message type, occupying one byte, and must be the first byte of a Bluetooth Data segment.

<bn> Display Brightness, occupying one byte, value range: 0 ~255

11.5 Set Switch Status

Command Description: This command is used to set the device's switch status.

Command Format:

Direction	Type	Params	Response
APP→LIGHT	0x05	<status>	<0x05><status>

Data Description:

<0x05> Message type, occupying one byte, and must be the first byte in a Bluetooth Data packet.

< status > Switch Status, occupying one byte, Value: 0- Off, 1- On

Acknowledgment:

<0x04> Message type, occupying one byte, and must be the first byte of a Bluetooth Data segment.

<status> Current switch status, occupying one byte, value: 0- Off, 1- On

11.6 Play the specified program list content

Command Description: This command enables playback of a specific content item within the program list. If the specified index does not correspond to any content, playback of the designated content will occur according to the configuration.

Command Format:

Direction	Type	Params	Response
APP→LIGHT	0x07	<index>	<0x07><index>

Data Description:

<0x07> Message type, occupying one byte, and must be the first byte of a Bluetooth Data segment.

<index> Program identifier, occupying one byte, value range: 0~8, 0xFF indicates auto loop playback of all programs, 0xFE indicates refresh and play once.

Acknowledgment:

<0x07> Message type, occupying one byte, and must be the first byte of a Bluetooth Data segment.

<index> Currently playing program list, occupying one byte, value range: 0~8, 0xFF indicates auto loop playback of all programs

11.7 Delete Specified Program List Entry

Command Description: This command deletes a specified entry in the Program List. If the deleted program is currently playing: 1. If auto-play is enabled, play the next program. 2. If auto-play is disabled, play the default content. 3. If all programs are deleted, play the default program.

Command Format:

Direction	Type	Params	Response
APP→LIGHT	0x08	<index>	<0x08><index>

Data Description:

<0x08> Message type, occupying one byte, always the first byte of a Bluetooth Data packet.

< index > The program number occupies one Byte, value range: 0~ 8 , 0xFF indicates deletion of all programs

Acknowledgment:

<0x08> Message type, occupying one byte, always the first byte of a Bluetooth Data packet.

<index> Program entry number, occupying one byte, valid values: 0~ 8, 0xFF indicates deletion of all programs.

11.8 Set Flip Status

Command Description: This command sets the flip status of the device's display content.

Command Format:

Direction	Type	Params	Response
APP→LIGHT	0x0C	<status>	<0x0C><status>

Data Description:

<0x0C> Message type, occupies one byte and is always the first byte of a Bluetooth Data segment.

<status> Flip status, occupies one byte, with values: 0- Normal, 1- XY Flip, 2- X Flip, 3- Y Flip

Acknowledgment:

<0x0C> Message type, occupies one byte and is always the first byte of a Bluetooth Data segment.

<status> Flip status, occupies one byte, with values: 0- Normal, 1- XY Flip, 2- X Flip, 3- Y Flip

11.9 Get Device Information

Command Description: This command retrieves various status information from the device.

Command Format:

Direction	Type	Params	Response
APP→LIGHT	0x1F	None	<0x1F><resp>

Data Description:

<0x1F> Message type, occupies one byte and is always the first byte of a Bluetooth Data segment.

Acknowledgment:

Parameter Name	Length	Description
<0x1F>	1byte	Message Type
< switch-s >	1byte	Device Switch Status, Value: 0- Off, 1- On

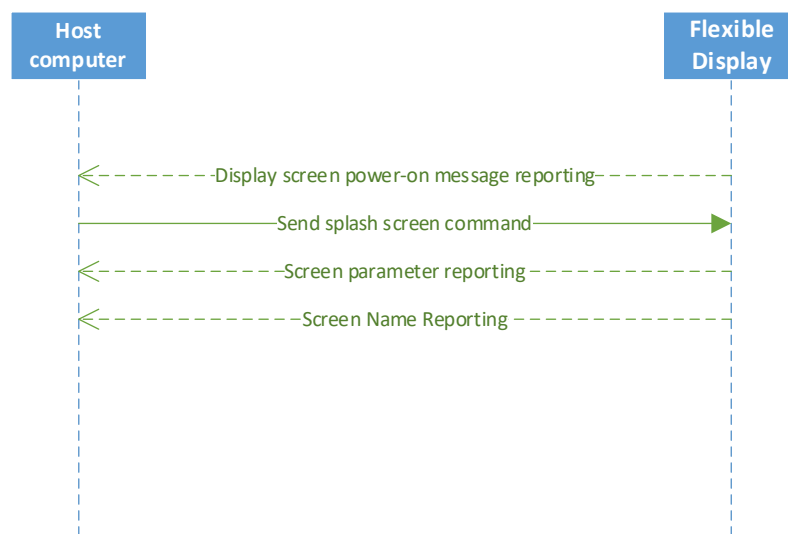
< bn >	1byte	Display Brightness, Value Range: 0~255
< flip >	1byte	Flip Status, Value: 0- Normal, 1- XY Flip, 2- X Flip, 3- Y Flip

Note: Please disregard any other returned fields.

12、 Boot Phase Protocol (Serial Port)

Commands during this phase correspond to interaction protocols for the screen boot process. Serial port implementations must handle this protocol segment and send commands in the specified sequence to ensure proper screen startup.

For BLE secondary development, this section of the protocol should be disregarded.



12.1 Screen Power-On Report

Command Description: This command is proactively reported by the screen to the host computer after power-on or reboot.

Command Format:

Direction	Type	Params	Response
LIGHTAPP→	0x32	<0x04>	<0x32><0x04><ack>

Data Description:

<0x32> Message type, occupies one byte and is always the first byte of a data segment.

<0x 04> Parameter, value: 0x04

Acknowledgment:

<0x32> Message type, occupies one byte and is always the first byte of a data segment.

<0x04> Parameter, value: 0x04

<ack> Response Status, 0 - parsing successful, others - parsing failed

Note: Upon receiving this command, if the host computer needs to start the screen, it must send the “Startup

Screen" command.

12.2 Screen Parameter Report

Command Description: After receiving the "Startup Screen" command from the host computer, the screen will report its parameters within a short time (approximately 50 ms).

Command Format:

Direction	Type	Params	Response
LIGHTAPP→	0x30	<ID-H><ID-L><RES-ROW><RES-C-H><RES-C-L><DEV-TYPE><DEV-VERSION>	<0x30><ack>

Data Description:

- <0x30> Message type, occupying one byte, and must be the first byte of a data segment.
- <ID-H> High byte of the device ID, occupying one byte
- <ID-L> Low byte of the device ID, occupying one byte
- <RES-ROW> Screen resolution row count, occupying one byte
- <RES-C-H> High byte of the screen resolution column count, occupying one byte
- <RES-C-L> Low byte of the screen resolution column count, occupying one byte
- <DEV-TYPE> Device type, occupying one byte, fixed value 0x02
- <DEV-VERSION> Device version number, occupying one byte, value range 0 to 255

Acknowledgment:

- <0x30> Message type, occupying one byte, and must be the first byte of a data segment.
- <ack> Response Status, 0 - parsing successful, others - parsing failed

Note: If the host computer does not require this parameter, it may be disregarded.

12.3 Screen Name Reporting

Command Description: After receiving the "Startup Screen" command from the host computer, the display screen will report back its screen name within a short period (approximately 50 ms). The host computer can use this name to differentiate between various products. The full-color display screen name is: C oolLEDU

Command Format:

Direction	Type	Params	Response
LIGHTAPP→	0x31	<name-arr>	<0x31><ack>

Data Description:

- <0x31> Message type, occupying one byte and always being the first byte of a data segment.
- <name-arr> Screen name, with a maximum length of 20 bytes.

Acknowledgment:

- <0x31> Message type, occupying one byte and always being the first byte of a data segment.
- <ack> Response Status, 0 - parsing successful, others - parsing failed

Note: If the host computer does not require the name, it may be disregarded.

12.4 Startup Screen

Command Description: This command initiates the screen startup. The screen will only power on correctly after the host computer sends this command.

Command Format:

Direction	Type	Params	Response
APP→LIGHT	0x23	<0x01><0x00>	<0x23><FLAG>

Data Description:

<0x23> Message type, occupying one byte, and must be the first byte of a data packet.

<0x01> Parameter, value: 0x 01

<0x00> Parameter, fixed value: 0x 00

Acknowledgment:

<0x23> Message type, occupying one byte, and must be the first byte of a data packet.

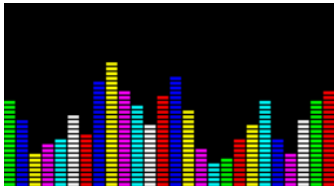
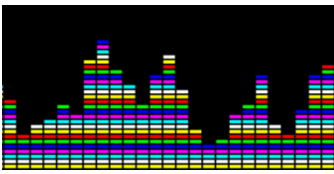
<FLAG> Flag, occupies one byte; the host computer does not need to consider the specific value of this variable.

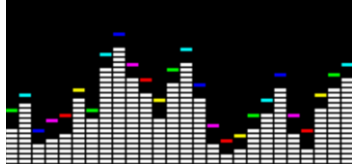
Note: After the device powers on, it will begin configuring screen-related parameters within a brief period.

13、 Appendix

13.1 Music Mode

Different Music Rhythm Effects, with values and corresponding effects as follows:

type	describe	Reference View
0x01	Seven-color vertical illumination towards one side; each column's color gradient from bottom to top follows the sequence red, yellow, green, cyan, blue, purple, red, with identical colors across each row.	
0x02	Seven-color vertical illumination towards both sides; each column's color gradient extends from the center towards both edges following the sequence red, yellow, green, cyan, blue, purple, red, with identical colors across each row.	
0x03	Seven-color vertical lighting toward one side; each column's colors transition from left to right in the sequence red, yellow, green, cyan, blue, purple, red, with each column uniformly colored	

0x04	Horizontal lighting from the center outward toward both sides; each column's colors transition from the center to the left and right in the sequence red, yellow, green, cyan, blue, purple, red, symmetrical on both sides, with each column uniformly colored	
0x05	Rhythm Mode with white falling points moving toward one side; falling point colors vary randomly	

13.2 Display Mode

Different display modes for Text or Graffiti:

Value	Mode Name	Remarks
1	Static	Content displays statically frame by frame in a loop
2	Continuous left shift	Content continuously moves left in a loop display
3	Continuous right shift	Content continuously moves right in a loop display
4	Up	Content continuously moves upward in a loop display
5	Down	Content continuously moves downward in a loop display
6	Stacking	Content stacks downward line by line
7	Unfolding Scroll	Content unfolds from the center toward both sides like a scroll in a loop display
8	Blinking	Content flashes once per frame in a loop display
9	Shift left	Content shifts left in a loop display, pausing for a specified duration after each full screen
10	Shift right	Content shifts left in a loop display, pausing for a specified duration after each full screen
11	Overlay left	
12	Right-side Overlay	
13	Horizontal Interleaving	

13.3 CRC32 Checksum Algorithm

To fully leverage the device hardware CRC 32 checksum capability, the host computer must use the identical checksum algorithm as the device. The CRC 32 checksum algorithm is as follows:



crc.js

14、 Example

14.1 Set Device Switch Status to ON

Set the Display Screen to ON

1. According to the protocol, the raw data is:

05 01

2. This data requires data formatting. After formatting (final transmitted data):

01 00 02 06 05 02 05 03

Field Description:

01 // Start Byte

00 02 06 // Data Length; since the second byte of data length is 02, mandatory conversion was applied

05 02 05 // Data; must be forcibly converted according to the rules

03 // End Byte

3. Device Response

01 00 02 06 05 02 05 03

This response data must be parsed in accordance with the data formatting protocol.

Parsed data is: 05 01